

①

Define matrix

Types of matrix

Define inverse of a matrix

Prove that if a square matrix is invertible then it is
(a) non singular, and (b) it has unique inverse.

Define Rank of a matrix

Det rank of the matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$,

Define Echelon Form.

Find the rank of a matrix by changing into E.F.

Define Normal Form.

Define Rank Nullity th.

Define L.D & L.I vector.

Show that $X_1 = (1, 1, 2)$, $X_2 = (1, 3, 5)$ and $X_3 = (5, 3, 4)$

are L.D. Also express vector as a linear combination of other two.

condition For consistency of the non homogeneous
system $AX=B$

examine for consistency the following eqⁿ and solve them
if they are consistent.

$$x+y+z=6, \quad 2x+y+3z=13, \quad 5x+2y+z=12, \quad 2x-3y-2z=-10$$

Find the rank of the matrix

(a) $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$;

(b) $\begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$

② For what values of λ & μ does the system of eqⁿ RPB
 $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+\lambda z=\mu$
 has (1) no solⁿ, (2) unique solⁿ, (3) infinite many solⁿ

For what values of k the eqⁿ.

$x+y+z=1$, $2x+y+4z=k$, $4x+y+10z=k^2$ have
 a solⁿ and solve them.

Define characteristic eqⁿ of a matrix

" " Eroot of a matrix.

" " Vector " " " "

Find the C.E of the matrix

$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ Also Find E.V and E.V.

State & Proof Cayley's Hamilton Theorem

Obtain the Characteristic eqⁿ of the matrix

$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$,

verify, that it's satisfied by A, and Hence find A^{-1}

Define differential Eqⁿ, Define
 order of D.E.

" degree of D.E.

Find degree and order of the differential eqⁿ

225-25

③

$$(D^3 + 1)y = 3 + e^x + 5e^{2x}$$

R.R.P.

$$(D^2 + D + 1)y = \sin 2x$$

$$\frac{d^2x}{dt^2} + 2n \cos x \frac{dx}{dt} + n^2 x = a \cos nt$$

such that $x=0$ and $\frac{dx}{dt}=0$, when $t=0$

$$(D-2)^2 y = 8x^2 e^{2x} \sin 2x$$

$$\text{Solve } x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$$

$$x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{(1-x)^2}$$

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$$

$$(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$$

$$\sin^2 x \frac{d^2 y}{dx^2} - 2y = 0 \quad (\text{I.F. } \cot x)$$

$$\frac{d^2 y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = \sin^2 x$$

$$\frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 1)y = -3e^{x^2} \sin 2x$$

$$\frac{d^2 y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = e^x \sec x$$

$$\cos x \frac{d^2 y}{dx^2} + 6 \sin x \frac{dy}{dx} - 2y \cos^3 x = 2 \cos^5 x$$

$$x^6 \frac{d^2 y}{dx^2} + 3x^5 \frac{dy}{dx} + a^2 y = \frac{1}{x^2}$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$$

$$\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x$$

$$\frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}$$

Solve using ~~set~~ series method

$$(-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

22-5-25

④ Solve $x dx + y dy = a^2 \left(\frac{x dy - y dx}{x^2 + y^2} \right)$

RRB

$$\left(y - \frac{x dy}{dx} \right) = a \left(y^2 + \frac{dy}{dx} \right)$$

$$\frac{dy}{dx} = \cos(x+y) + \sin(x+y)$$

Solve $\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y - \left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} \frac{x dy}{dx} = 0$

Solve $x^2 y dx - (x^3 + y^3) dy = 0$

Solve $\frac{dy}{dx} = \frac{(x+y-1)^2}{4(x-2)}$

Solve $\frac{dy}{dx} = \frac{x-y-2}{2x-2y-3}$

Define linear differential Eqⁿ or Bernoulli's Eqⁿ

Solve $(x + 2y^3) \frac{dy}{dx} = y$

" $(1+y^2) dx = (\tan^{-1} y - x) dy$

Solve $\sin y \frac{dy}{dx} = \cos y (1 - x \cos y)$

" $\frac{dy}{dx} = \frac{1}{xy(x^2 y^2 + 1)}$

Condition for exactness

Solve $(1 + e^{x/y}) dx + e^{x/y} \left(1 - \frac{x}{y}\right) dy = 0$

Solve $(x^2 y^2 + xy + 1) y dx + (x^2 y^2 - xy + 1) x dy = 0$

Solve $(x^4 y^4 + x^2 y^2 + xy) y dx + (x^4 y^4 - x^2 y^2 + xy) x dy = 0$

Solve $\frac{d^4 y}{dx^4} + a^4 y = 0$

$$\frac{d^3 y}{dx^3} - 9 \frac{d^2 y}{dx^2} + 26 \frac{dy}{dx} - 24 y = 0$$

$$(D^2 - 2D + 4) y = e^x \cos x$$

$$(D^2 + 2D + 1) y = x \cos x$$

$$\frac{d^2 y}{dx^2} + a^2 y = \sec ax$$

22-5-25

RRB =

⑥ Define P.D.E.

Solve $z - xp - yq = a\sqrt{x^2 + y^2 + z^2}$

Solve $p + 3q = 5z + \tan(y - 3x)$

Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$

Solve By Charpit's method

$$2xz - px^2 - 2qxy + pq = 0$$

$$(p^2 + q^2)y = qz$$

Define wave Eqⁿ, Heat Eqⁿ, Laplace Eqⁿ.

Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, by the method of separation

of variable given that

$$u(x, 0) = 6e^{-3x}$$

Solve one dimensional Heat Eqⁿ.

Solve one dimensional wave Eqⁿ

Best of luck

22/5/25